MATH 3312, SPRING 2026: HOMEWORK 4

(1) Determine if the following statements are true or false. Give a line or two in justification.
  (a) The fields $\mathbb{F}_5[x]/(x^4+x+1)$ and $\mathbb{F}_5[x]/(x^4+x^2+x+1)$ are both isomorphic to each other. (*You can assume the given polynomials are irreducible in* $\mathbb{F}_5[x]$)
  (b) If k is a finite field of order 32, then $x^{64}-x$ has 32 solutions in k.
  (c) There are exactly 36 irreducible polynomials of degree 4 in $\mathbb{F}_3[x]$.
  (d) There is an irreducible cubic polynomial in $\mathbb{F}_5[x]$ with exactly one zero in a field of order 125.
  (e) Every field of order 81 contains a field of order 9.

(2) Suppose that $f(x) \in \mathbb{F}_p[x]$ is an irreducible polynomial of degree r and let k be a finite field of characteristic p. Show that the following are equivalent:
  (a) $|k| = p^d$ where $r \mid d$;
  (b) $f(x)$ admits a zero in k.
  Hint: This is essentially a reformulation of problem 4(b) and (a)⇒(b) in problem 5 of Homework 3.

  The next problem completes the proof of (b)⇒(a) of problem 5 of Homework 3.

(3) Let $f(x) \in \mathbb{F}_p[x]$ once again be an irreducible polynomial of degree r, and suppose that $f(x) \mid x^{p^d}-x$. Set $k = \mathbb{F}_p[x]/(f(x))$ and $\alpha = x + (f(x)) \in k$.
  (a) Show that r is the smallest positive integer m such that $\mathrm{Fr}^r(\alpha) = \alpha$
  Hint: Use the previous problem.
  (b) Show that $\mathrm{Fr}^{\gcd(r,d)}(\alpha) = \alpha$.
  (c) Conclude that $r \mid d$.

  The previous problem completes our proof of Galois' Theorem:

Theorem 1. *For every prime p and every $d \geq 1$, there is a* unique *field of order p^d up to isomorphism.*

Notation 1. We will denote the unique field of order p^d by $\mathbb{F}_{p^d}$.

Caution 1. When $d \geq 2$, this is *very* different from $\mathbb{Z}/p^d\mathbb{Z}$. Do *not* confuse the two.

(4) Suppose that we have $\alpha \in \mathbb{F}_{p^d}$ and that $m_\alpha(x) \in \mathbb{F}_p[x]$ is its minimal polynomial over $\mathbb{F}_p$. Set $r = \deg m_\alpha(x)$.
  (a) Show that $\mathrm{Fr}(\alpha) = \alpha^p$ also has minimal polynomial $m_\alpha(x)$.
  (b) In fact, show that the set
$$\{\mathrm{Fr}^m(\alpha) : \ 0 \leq m \leq r-1\} \subset \mathbb{F}_{p^d}$$
  is exactly the set of r distinct zeros of $m_\alpha(x)$ in $\mathbb{F}_{p^d}$.

(5) Show that the following are equivalent for $\alpha \in \mathbb{F}_{p^d}$ and an integer $r \geq 1$:
  (a) r is the smallest positive integer such that $\mathrm{Fr}^r(\alpha) = \alpha$;
  (b) The smallest subfield of $\mathbb{F}_{p^d}$ containing α has order p^r;
  (c) $m_\alpha(x)$ has degree r.

(6) Suppose that $\alpha \in \mathbb{F}_{p^d}^\times$ is an element of order m. Show that

$$\deg m_\alpha(x) = \text{order of } p \text{ in } (\mathbb{Z}/m\mathbb{Z})^\times.$$

Note the reversal of the roles of p and m! This is an example of what is called a reciprocity law.

Definition 1. Let $\mathrm{Aut}(\mathbb{F}_{p^d}/\mathbb{F}_p)$ be the set of field of automorphisms of $\mathbb{F}_{p^d}$. This is usually also denoted $\mathrm{Gal}(\mathbb{F}_{p^d}/\mathbb{F}_p)$, which is notation we will come back to later in the semester. This is a *group* under composition (why?).

Can you guess what Gal *stands for?*

(7) (a) Show that there is a unique group homomorphism $\mathbb{Z}/d\mathbb{Z} \to \mathrm{Aut}(\mathbb{F}_{p^d}/\mathbb{F}_p)$ that takes 1 to Fr;
(b) Show that this gives a group action of $\mathbb{Z}/d\mathbb{Z}$ on $\mathbb{F}_{p^d}$ for which we obtain a bijection

$$\mathrm{Irr}_{|d}(\mathbb{F}_p) \xrightarrow{\simeq} \{\text{Orbits of the action of } \mathbb{Z}/d\mathbb{Z} \text{ on } \mathbb{F}_{p^d}\}$$

We will see later in the term that the homomorphism in (a) is in fact an isomorphism.

(8) Combine the previous problem with Burnside's Lemma to show that, if N_r is the number of irreducible polynomials in $\mathbb{F}_p[x]$ of degree r with leading coefficient 1, then we have

$$\sum_{r|d} N_r = \frac{1}{d}\sum_{m=0}^{d-1} p^{\gcd(m,d)}.$$